

Khoa Thai Dang Tran

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PROFESSIONAL SUMMARY

Software engineer and MS Computer Science student (graduating December 2026) with 2+ years of professional full-stack experience building production systems in Java Spring Boot, JavaScript, Vue.js, and PostgreSQL for major Japanese retail chains. Shipped LLM-integrated features and write tests at every level (unit, integration, end-to-end) wired into CI.

EDUCATION

University of the Pacific, Stockton, CA

January 2025 - Present

MS Computer Science | GPA: 3.76 | Expected graduation: December 2026

University of Science, VNU - HCM, Vietnam

2018 - 2022

BS Information Systems | Graduated: 2022

EXPERIENCE

Software Engineer

November 2022 - December 2024

UNICCS Co., Ltd

- Architected and delivered full-stack features for TAURUS, a core retail ERP covering purchase ordering, SKU-level inventory, and departmental P&L, used by major Japanese chains including Don Quijote, Welcia, and Sundrug, built with Java Spring Boot and responsive frontends in JavaScript and Vue.js.
- Reduced query response times by 40% through SQL tuning, indexing, and schema redesign on PostgreSQL and DB2, supporting high-volume transaction processing.
- Cut a dashboard stored procedure's runtime from 10 minutes to 30 seconds by isolating the slow join and indexing the key field.
- Deployed and maintained services on AWS, automating build, deployment, and monitoring workflows with Shell scripting to cut manual release effort.
- Debugged and resolved production issues reported by client teams, tracing defects across the Spring Boot backend and Vue.js frontend.

Tech: Java Spring Boot, JavaScript, Vue.js, PostgreSQL, DB2, SQL, AWS, Shell, HTML, CSS

Software Engineer Intern

December 2021 - June 2022

FPT Telecom

- Built RESTful APIs with Node.js and Express powering POP management dashboards for real-time monitoring of telecom infrastructure across multiple POP stations.
- Developed interactive React and Next.js dashboards with Highcharts visualizations, backed by MongoDB schemas designed for efficient querying of monitoring data.

Tech: React.js, Next.js, Node.js, Express, MongoDB, Highcharts

SKILLS

- Languages:** JavaScript, TypeScript, Python, Java, C++, C#, SQL, HTML, CSS
- Frameworks:** React.js, Next.js, Vue.js, Node.js, Express, Spring Boot, Tailwind CSS
- Databases:** PostgreSQL, MySQL, MongoDB, DB2, SQL Server, Oracle
- Testing & CI/CD:** Vitest, Playwright, unit testing, integration testing, end-to-end testing, GitHub Actions
- Cloud & Tools:** AWS, Git, GitHub, Shell scripting, Jupyter Notebook, Claude Code (AI-assisted development)
- Concepts:** Object-Oriented Programming (OOP), Data Structures and Algorithms, RESTful APIs, Database Design, Agile/Scrum, Machine Learning, NLP, LLM Integration, Prompt Engineering, AI Agents

PROJECTS

CineVibes — Personal Project

[Live Demo](#) · [Source Code](#)

- Built a full-stack movie discussion platform with spoiler-aware comment threads, trending rankings, and CineBot, an in-app AI assistant (Groq-hosted LLM) grounded in the app's MongoDB catalog through a search tool.
- Wrote 57 automated tests: 51 Vitest API and integration tests across 4 suites, plus 6 Playwright end-to-end flows, running alongside lint and build on every push and pull request through GitHub Actions.

Tech: React, Node.js, Express, MongoDB, Vite, JWT Auth, Groq, Vitest, Playwright, GitHub Actions

ML-Assisted Natural Language to SQL Query System — Personal Project

[Live Demo](#) · [Source Code](#)

- Built a CSV-agnostic natural language to SQL system: users upload any CSV and ask questions in plain English. Pairs a BERT intent classifier fine-tuned on 1,250 labeled examples spanning 14 domains and 5 query intents with a self-built, schema-aware SQL generator and no external LLM API calls.

Tech: Python, BERT (bert-base-uncased), PyTorch, Streamlit, SQLite, Pandas, Matplotlib, Seaborn

ExpenseTracker — Personal Project

[Live Demo](#) · [Source Code](#)

- Developed a client-only expense tracker (filtering, budgets, dashboard analytics, CSV/JSON/PDF export across 4 report types) end-to-end with Claude Code: compared three parallel implementations of the export feature, fixed a CSV-injection vulnerability the comparison surfaced, and covered the aggregation modules with 31 unit tests.

Tech: Next.js, React, TypeScript, Tailwind CSS, Recharts, jsPDF, Vitest